

## Question Bank

### Electricity

#### **Lecture – 1:**

1. What is electrostatics?
2. What is lightning?
3. How can you stay safe during a lightning storm?
4. Explain the term 'Charge is quantized'.
5. State and discuss Coulomb's law. Write down the vector form of it.
6. What are limitations of Coulomb's law?
7. Define a Coulomb.
8. Define relative permittivity.
9. Solve the mathematical problems from lecture sheet.

#### **Lecture – 2:**

1. What is electric field?
2. Define "electric field intensity".
3. What do you understand by electric lines of force?
4. Draw the electric lines of force for two nearest opposite charges?
5. Write down the equation of electric field for point charge.
6. Write down the equation of electric field for the system of charges?
7. Derive the equation of electric field due to a long straight charged wire.  
Or, Show that " the electric field due to a long straight charged wire is inversely proportional to the perpendicular distance of the point from the wire".
8. Solve the mathematical problems given in the lecture sheet.

#### **Lecture – 3:**

1. What is the electric potential? Mention the unit of it.
2. Distinguish electric potential and electric potential energy.
3. Write a short note on equipotential surface.
4. Define electric field from potential gradient.
5. Solve the mathematical problems given in the lecture sheet.

#### **Lecture – 4:**

1. What is electric current & current density? Give the unit of them.
2. Construct a relationship between current density & charge density.
3. Define drift velocity.
4. Show that, "drift velocity is inversely proportional to the number of charge per unit volume".
5. Define the resistivity from the law of resistances.
6. What is conductivity? Relate it with resistivity.
7. State the ohm's law. Convert the ohms law to the another form.
8. Write a short note about "resistivity in an atomic view".
9. Solve the mathematical problems given in the lecture sheet.

#### **Lecture – 5:**

1. Define Electric flux. What is the unit of it?
2. State the Gauss Law with mathematical expression.
3. Is it possible to derive Coulomb's law from Gauss law? – justify your answer.

#### **Lecture – 6:**

1. What is capacitor & capacitance?
2. Define 1 Farad.
3. Draw the parallel plate capacitor and formulate an equation of it?
4. Explain the capacitance of cylindrical capacitor is independent of charge of the capacitor and the potential difference between the plates of the capacitor.
5. Is it possible to derive an equation for the capacitance of a spherical capacitor by using Gauss law? – Explain it.
6. Practice the mathematical problems given in the lecture sheet.

**Lecture – 7:**

1. Explain the derivation of equation of electric field due to a long straight charged wire by using Gauss law.
2. Show that “the net flux of a cylindrical Gaussian surface placed in an external electric field is zero”.
3. Define Dielectric Constant.
4. What is the effect of dielectric materials in capacitor?

**-: The End :-**